GemeindeSim: a bilingual civic what-if simulator on Apertus 1.5

Hack Apertus 2026 (online) · Track 2B, own project · Team **1ne** (repository owner: hatif03) · 5–15 Oct 2026 · code: github.com/hatif03/gemeindesim · demo videos: youtu.be/ZdXAJjCUhZI (2 min), youtu.be/BZnqYIDV63Q (full run) · reference numbers [n] point to the list after section 8

1. Summary

GemeindeSim is a what-if and pre-test tool for official voting texts. It reads a Swiss municipal vote in German and French and lets a fictional town of up to 25 residents, played by Apertus 1.5 [61], read it, talk, argue with citations and form a stance on a pixel-art map; a closing report says *who is affected and why*, never how to vote. It follows the idea of populated prototypes [2] and generative agents [1]; we found nothing comparable for Swiss voting texts (section 1.2). We also used the town as a test bench for Apertus (8B and 70B, two endpoints, more than 20,000 logged calls and trials in 32 experiments). **Headline result:** Apertus is an excellent *speaker* (112/112 resident lines in the right language; it abstains when a premise is missing) and an unreliable *decider*: asked for a stance, all 15 residents said yes, even with no persona, and on 54 real Swiss votes the simulated electorate was 34–37 points too favourable, as LLM respondents elsewhere lose variance, flatten groups and drift to the centre [14, 15, 31, 32]. So **the model voices; the application owns the stance, the arithmetic and the sources** [35–37]. Same seeds, before and after: valid citations 0/159 → 100 %, reports asserting an outcome 12/30 → 0/30, residents whose words match their stance ≈0.5 → ≈0.9; on a real 48-page Federal Council booklet [60] 12/14 questions were answered correctly and every unanswerable one was declined. It is a what-if explainer, not a vote predictor [29, 34, 39].

1.1 Why it matters: the Swiss evidence

Swiss voters decide on written text, and the text can be wrong or misread. In 2019 the Federal Supreme Court annulled the 2016 *Heiratsstrafe* vote because the booklet said about 80,000 couples were affected when 454,000 were [50]; a 2023 parliamentary audit found the booklet's four-eyes check is not always done by people with the right expertise [51]; 43 % of 1,513 respondents could not say what the November 2018 vote was about [52]. In many of the 2,110 municipalities the assembly sees 4–8 % of eligible voters, so the written proposal is what most people read [58]; the content of a booklet moves votes (an indicative, commissioned study [54]) and many voters feel they understood a proposal they did not [57]. The Federal Chancellery edits and reviews the booklet by hand and, as far as we found, tests only its layout with voters, not its wording [51, 60]. A municipal booklet must give objective and complete explanations [53], yet the Federal Chancellery resists plain-language rewrites for fear of unlawful influence [55], so a pre-test has to check wording rather than simplify it, and Swiss voters accept AI in authority communication only with human oversight [56]. We found no actor that pre-tests the wording of an official text with simulated readers before it is published; the nearest Swiss work measures the neutrality of booklets after publication [20] or explains votes to voters [59].

1.2 Similar applications, and what they are used for

Simulated people and AI-assisted listening are used in research and, increasingly, in government and market research.

Application	What it is used for	What is known about it
Singapore GovTech *Echo*, Sept 2026 [44]	AI personas, built from past focus-group transcripts, react to a draft government announcement before it is released	the closest government case; early testing, still validated against real focus groups; "personas generate hypotheses, not conclusions"
Social Simulacra, Stanford, 2022 [2]	an LLM generates thousands of community members so designers can test a community design before launch	the origin of the "populated prototype" idea we follow
Generative agents and the 1,052-person study, Stanford, 2023 and 2024 [1, 3]	agents with memory, reflection and interview-grounded personas simulate individuals and small societies	match survey answers at 85 % of participants' own two-week consistency; the authors' spin-off sells it (company claims only)
Predicting experiment results, *Nature* 2026 [13]	GPT-4 simulates 70 US survey experiments to pre-test messages	direction and rank right (r = 0.85), effect sizes systematically too large: good for comparing, not for levels
AI-mediated deliberation: Habermas Machine, DeepMind 2024 [28]; Consult, UK government 2025 [45]	write group statements that people prefer 56 % of the time; sort 2,000+ consultation answers (F1 0.76); practitioners argue for simulated deliberation as a complement to human deliberation [27]	they work on real people's words, not simulated ones; no public deployment of the first
Synthetic respondents in market research: Aaru, Electric Twin, Ipsos, Qualtrics [46]	persona panels to test ads, products and campaign messages	accuracy figures are the vendors' own; the independent test (Pew 2026 [40]) found 11–13 points average error
Human pre-testing of ballot wording: UK Electoral Commission 2012 and 2015 [47]; citizen panels in Oregon and Sion (VS) [48]	203 interviews and 10 focus groups changed the wording of the Scottish referendum question; panels write the voter-pamphlet statement	the gold standard GemeindeSim is a cheap first pass for: it costs weeks and thousands of francs per question

Beyond the table: large agent simulations use small open models for cost and reproducibility [4–6], and whether LLM social simulation is a sound method is debated [7, 35]. LLM respondents have been studied for survey experiments [13, 25], German and European vote choice [21, 22, 24], Swiss politics [15–19], augmented democracy [23], news-feed design [11] and policy-impact scenarios [26, 30], with warnings about how they represent groups [33]; a US town used AI to cluster its residents' own ideas [49].

2. Architecture

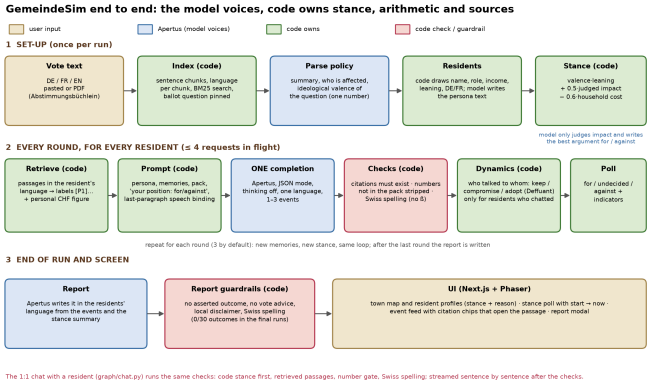


Fig. 1. End to end. Blue: Apertus voices. Green: code owns. Red: code checks the model. Beige: user and screen.

Stack. Two Docker services. A FastAPI + LangGraph backend holds the standard and a two-phase (initiator/reactor) round graph; sentence-level chunking with BM25 retrieval [62] (per-chunk language, the ballot question pinned) and an optional hybrid with a multilingual dense model [63] fused by reciprocal rank [64]; an async model client with an adaptive concurrency limiter, retries and sentence-level streaming; a grounded 1:1 chat whose answers cite their passages [65]; token accounting; the report writer with its guardrails; a Socket.IO stream; and a multi-seed spread run. A Next.js + Phaser frontend shows the map, the event feed with citation chips, the stance poll, the influence graph, the report, a run-metrics panel and a recorder/loader for saved runs. The corpus is read-only; the only outbound call is the OpenAI-compatible `LLM_BASE_URL`. No personal data is used: residents are generated at run time (code draws name, role, income band, leaning and language; the model writes the persona text).

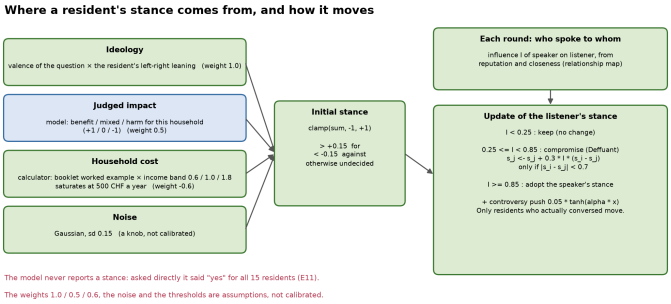


Fig. 2. Stance is computed in code (weights are assumptions, not calibrated).

Code-owned controls (each added after a measured failure, section 5.3):

- **Stance**: ideology × leaning + 0.5 × judged impact − 0.6 × computed household cost + noise (sd 0.15); thresholds ±0.15 (persona text alone explains little of LLM answers [43]).
- **Opinion dynamics**: bounded confidence (Deffuant [8], fitted to data in [9]): keep / compromise / adopt by influence; only residents who conversed move. LLM agents otherwise drift to consensus or polarise by prompt [10, 38].
- **Sources**: passages carry labels [P#] and every claim must cite one, as in citation-grounded generation [65]; a citation that does not exist is dropped; the ballot question is always in the prompt.
- **Numbers**: numeral gate strips figures not in the pack or the calculator; household CHF comes from a calculator on the booklet's worked example.
- **Speech binding**: last paragraph of the prompt states the code stance ("your position: for/against").
- **Report guardrails**: no asserted outcome, no vote advice, Swiss spelling (no ß), a disclaimer that the town is fictional and not a poll.

2.1 Target architecture and dependencies (mandatory)

Deployment target: (c) Swiss sovereign cloud. The application (two containers) runs on any Docker host and calls a Swiss-hosted OpenAI-compatible endpoint: the CSCS inference API (the judging endpoint) or the hackathon gateway. One build, three variables (`LLM_NAME`, `LLM_BASE_URL`, `LLM_API_KEY`); model ids are mapped between the two endpoints' styles and concurrency adapts (`LLM_CONCURRENCY=auto`). **(a) On-premise and (b) air-gapped** are configuration paths: point `LLM_BASE_URL` at an OpenAI-compatible server hosting open weights, e.g. `vllm serve swiss-ai/Apertus-v1.5-8B --port 8001` on a GPU host with `LLM_BASE_URL=http://host.docker.internal:8001/v1`; the 8B was about 4× faster per simulation than the 70B. **Honest status:** every behaviour was measured on two *hosted* endpoints; no GPU was available, so no local run was measured (a probe script runs the same capability table against any endpoint).

External dependencies. *Build time:* Docker base images; Python packages (FastAPI, LangGraph, the LangChain OpenAI client, Socket.IO, Pydantic, pypdf) and npm packages (Next.js, Phaser) from the public registries; Google-hosted web fonts fetched by the frontend build; optionally the 2.2 GB multilingual-e5 model [63] for hybrid retrieval. *Run time:* the backend makes one kind of outbound call, to the endpoint; the corpus is read-only inside the container; the browser page also loads one web font (IBM Plex Mono) from Google Fonts in its ASCII-art component, which an air-gapped installation would have to bundle (not yet done).

3. Use of Apertus

Model and role. Apertus 1.5 [61]: the 70B for the town, the 8B as the fast alternative and as the fallback after a timeout. Policy parsing, personas, pro/con arguments, resident turns, impact judgement and the report: all with the 70B, in `json_object` mode, thinking off, one language per completion, with an *instruction-placeholder* example in the resident's language (49/49 first-attempt valid, 0 % parroting; a filled example is parroted, 8B 53 %; strict `json_schema` breaks the 8B). DE ↔ FR translation of lines is a separate call; the 1:1 chat is streamed per sentence. The model is used for inference only (no fine-tuning, adapters or quantisation

by us); it is served by the endpoint (the Swiss AI Charter alignment is Apertus's own, not ours). Open models are what the large agent simulators use too [4–6]; fine-tuning Swiss models on viewpoints is the route of [19]. Retrieval, numbers, stance and dynamics are plain code: no tool calls (parallel calls fail 11/11; thinking plus JSON silently skips the reasoning).

4. Data

Sources and licences. (1) Fictional texts written for this project: a bilingual Steuerfuss and school-credit vote (with and without the referendum committee's arguments), six more fictional situations and an English tariff text; CC-BY-4.0 with the code; no real municipality or person. (2) Swissvotes, University of Bern [59]: titles, dates and results of 54 federal votes 2021–2026, used only to evaluate (E8), cited to the source. (3) One real Federal Council booklet [60] (28 Sept 2025, 48 pages), a public official text, downloaded by a script for evaluation and **not stored in the repository**. **Personal data and consent.** None: residents are generated at run time, no human subjects took part, and no personal or non-redistributable data is in the repository (its data folder is a few kilobytes).

5. Evaluation

5.1 Task, metric and baseline

The task is a grounded, bilingual town simulation of a vote text that never advises. Metrics, each measured on logged calls: groundedness (cited passage ids that exist; figures found in the text or computed), report safety (reports asserting an outcome; vote advice), stance behaviour (spread of the starting poll; agreement of words with the code stance), real-world sanity (yes-share and rank correlation against 54 real federal votes), retrieval (recall@4 on 14 gold questions of a real booklet) and cost and speed (tokens, latency, requests in flight). The **baseline** is the inherited pipeline before our changes, on identical seeds: section 5.3 is the baseline-versus-final comparison table, section 5.2 the limits we measured in Apertus itself, section 5.5 the index of all experiments.

5.2 Limits of Apertus 1.5 in an agent loop (hosted gateway, 5 Oct; replicated on CSCS, 6 Oct)

Behaviour	Result
Parallel tool calls	fail in 11/11 trials per model, also with `tool_choice="required"`; the 70B writes the second call as text
Thinking vs JSON mode	with `json_object` no reasoning span (16/16); the 70B answers a trap puzzle wrongly in 15 tokens; thinking alone 6/8 (70B), 8/8 (8B)
`temperature 0`	5 identical prompts → 3 (70B) and 5 (8B) distinct outputs ("deterministic" settings are not deterministic in other LLMs either [43]); `seed` has no effect; the same on CSCS (2.7 distinct of 3)
Swiss facts, 15 questions	70B 12/15, 8B 9/15 (CSCS 11/15, 9/15); both define *Steuerfuss* wrongly; 4,000 × 6 % → 280 (70B), 360 / 72 (8B), correct 240
Language, 112 events	100 % in the resident's language; the 70B writes ß at 2.6 per 1,000 characters (Swiss German has none: filtered in code)
Long context	needle recall 60/60 up to ≈105 k tokens (70B ≈36 s per call there); CSCS to the advertised 262 k

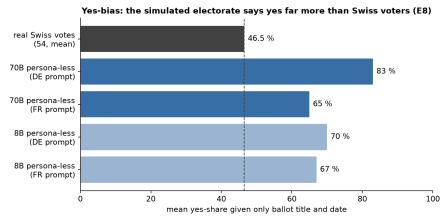


Fig. 3. The yes-bias: asked directly, every resident says yes (compare the loss of variance in [14, 32]).

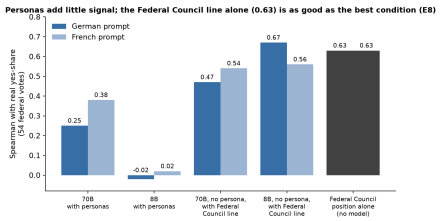


Fig. 4. 54 real Swiss federal votes [59]: bias and weak signal (compare [15, 16, 21]).

Test	Result
Stance asked directly (15 residents)	**15/15 yes** with full persona, role-only and no persona (E11); 14/15 with a balanced booklet; impact judged "benefit" for 14/15; on CSCS the 70B, 8B and 70B-thinking still say 14 yes / 1 no with a persona and 15 yes without (E24; the 8B-thinking model left answers unparsed)
54 real federal votes, ballot title only (E8)	simulated yes-share 83 % (DE prompt) / 65 % (FR) vs real 46.5 %; Spearman with real yes-share 0.25–0.38 (70B with personas), ≈0 (8B); left-right self-placement removes the signal; the Federal Council position alone: 0.63, direction accuracy 0.75; replicated on CSCS within 0.01; model and language effects were also found on Swiss referenda [15, 16] and in polls [40, 41]
Authority cue	shown the recommendation, the 8B follows it on 98 % of votes (70B 83 %), a form of the sycophancy described for LLM debates [38]; German prompts 19 points more favourable than French (prompt language changes answers: [16, 21, 42])
Left to itself at the end (E13)	the model agreed with the code stance for 40 % of residents and turned 3 of 4 opponents into "yes" → the stance stays in code

Conclusion: a what-if explainer, not a vote predictor (the independent evidence agrees: 11–13 points average error for synthetic polls [40], a model choice that changes a by-election call [41]). It compares conditions (with/without an official recommendation, German vs French, a counter-argument added); it does not give absolute shares.

5.3 Baseline versus final: defects found beyond the model, and the fix (identical seeds)

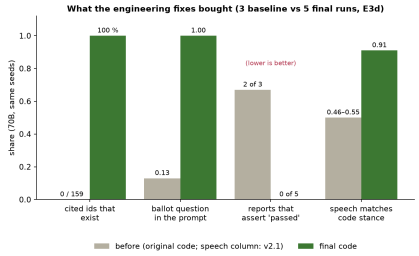


Fig. 5. Before / after (3 baseline vs 5 final 70B runs, same seeds).

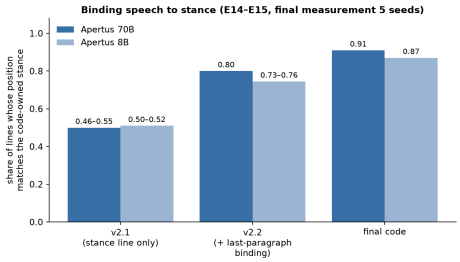


Fig. 6. Speech binding: residents' words match the code stance.

Defect	Before	After
Cited source ids that exist (cf. [65])	0 / 159	100 % (16 runs)
Reports asserting “the measure passed” / mixing German and English	12/30 / 5/5 runs	0/30 / 0/16 runs
Chats opening with a self-introduction (70B)	27 %	0 % (metric corrected, C1)
Polarisation of a town where nobody talks (15 rounds)	|x| 0.51 → 0.69	0 (drift needs a conversation)
Residents' words match their stance (70B / 8B)	0.46–0.55 / 0.50–0.52	0.88–0.91 / 0.85–0.87
Undecided residents sounding undecided (70B)	0.47	0.74 (8B 0.53 → 0.68)
`make run` on a clean checkout	failed (no Linux native packages in either lockfile)	builds; verified again from a fresh clone of the public repo on 13 Oct

Real booklet and spread across seeds. On the 48-page Federal Council booklet of 28 Sept 2025 [60] (83 k characters, 209 chunks, two measures) the retriever puts the gold evidence into four passages for 11/14 questions (old retriever 10/14); the 70B answered 12/14 correctly (hand-graded), abstained on 3/3 unanswerable questions and abstained wrongly only where the evidence was not retrieved: *retrieval, not the model, is the bottleneck*. Personas and the stance prior are drawn per seed and `T=0` is not reproducible, so one run is not a result: across five 70B seeds the final share *for* ranges 0.00–0.80 (8B 0.00–0.40), and the poll did not change in 4 of 5 runs within three rounds (small design changes can move simulation outcomes by tens of points [36]). Compare-conditions (E19, 8B, three paired seeds): adding the referendum committee's counter-arguments lowers the mean stance in all three seeds (−0.28); a yes/no council recommendation does not move it consistently, because the stance never reads it.

5.4 Second endpoint, engineering checks and the demo run

CSCS (E20–E28). Capability matrix, throughput, long context, determinism, structured output, stance on four models, retrieve-first vs whole-booklet, the final code on 5 seeds × 2 sizes, and the grounded chat live. All v2 fixes replicate (valid citations 1.00, no asserted outcome, speech matches the code stance 0.85–0.88), and so do the Swiss-knowledge and real-vote results. Unchanged: the yes-bias (E24), one tool call, non-reproducible `T=0`. With the complete booklet in the chat prompt (6,000–100,000 characters) the real revenue question is answered 5/5 instead of 0/5. **One build, both endpoints (E31; same code and seeds, the 5-resident 3-round run took 61 s on CSCS and 126 s on the gateway; the gateway answers 429 above 4–5 requests in flight, CSCS took 2,496 requests up to 96 in flight without one; Python used 2.2–5.0 % of the wall time, E29):** ids are mapped and the limiter adapts (three iterations): 25 residents × 2 rounds took 89–97 s on CSCS and 654 s on the busy gateway.

Mentor questions, answered with measurements (E29–E32)

- **Python, async, context.** The wall time is the longest chain of dependent calls at ≈60 tokens/s; every call is async; prompts use at most 1.1 % of the 262 k window, so no compaction is needed; 17–25 % of prompt tokens were cached on CSCS (moving the instruction block to the front would save ≈0.3 s per call but moves the speech-binding paragraph: not done).
- **Streaming, tokens, subagents.** The chat streams sentence by sentence after the number gate and citation check (first text 1.5 s, whole answer 2.4 s); JSON turns and the report are not streamed (they must be complete to be validated). Tokens are counted from the endpoint's `usage` and shown in a run-metrics panel. Each resident is an isolated context run in parallel; there is no tool loop.

RAG against the usual baselines (E30; 14 gold questions, 209 chunks of the real booklet [60]). recall@4 / MRR@10: our BM25 [62] 0.79 / 0.29; dense multilingual-e5-large [63] 1.00 / 0.89; **hybrid BM25 + e5-large (reciprocal rank fusion [64]) 1.00 / 0.91**; a small paraphrase model 0.71 / 0.56 (worse than BM25). n = 14 on one German document: one-question differences are noise. The hybrid is an option, off by default because it needs a 2.2 GB local model that an air-gapped deployment would have to ship.

The demo run (recorded from the running app on CSCS)

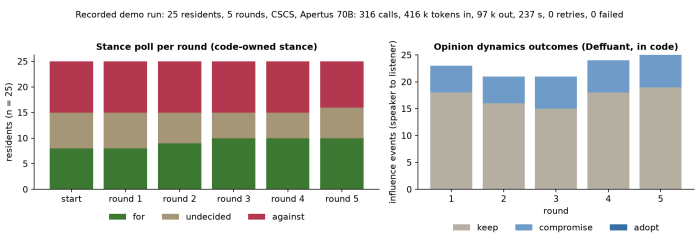


Fig. 7. 25 residents (13 DE, 12 FR), 5 rounds, Apertus 70B on CSCS, balanced Linden text, standard graph.

316 calls, 416 k tokens in and 97 k out (25 % cached), mean latency 9.0 s, p95 24.6 s, 0 retries, 0 failures, 237 s wall (limiter 4 → 32). All 150 resident dialogue lines were grounded with 0 invalid citations; the household figures 144 and 432 CHF are the calculator's.

The poll moved from 8 for / 10 against / 7 undecided to 10 / 9 / 6: opinions move a little, as in the five-seed study; the report names who is affected and ends with "no voting recommendation". Three candidate runs were recorded (25×5 twice, 25×4); the one with the most mixed start was kept. The backend caps residents at 25.

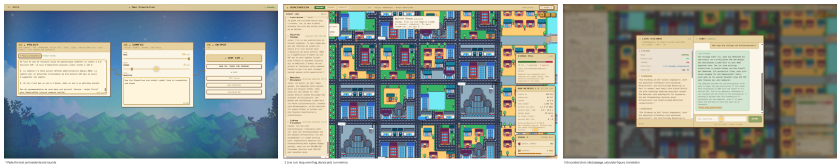


Fig. 8. The app: editor (paste, residents, rounds), a live run with the stance poll and the run-metrics panel (tokens, cache share, latency, in-flight limit, retries), and a grounded resident chat.

5.5 Experiment index (every call is logged; the scripts and raw logs are in the repository)

ID	Question	Setup	Result
E0	Is the offline test suite really offline, and how long does it take?	69 offline tests, two runs, timed	69 passed; first run 420 s, second 50 s. One end-to-end test (45 s) calls the live gateway, so the suite is not offline.
E1	Do the earlier claims on tool calls, thinking and determinism replicate? (determinism cf. [43])	70B and 8B; n = 11 (3 at T=0, 8 at T=0.7); thinking n = 8	Parallel tool calls fail 11/11 on both models. Thinking + json_object gives wrong 0.10 (no span, 16/16). T=0 not deterministic: 70B 3, 8B 5 distinct outputs of 5.
E2/E2b	Which example and format conditions make resident JSON valid and diverse?	49 logged resident prompts, 70B and 8B, 4 conditions; first run void (429s)	response_format hypothesis refuted (strict JSON 100% without it). Filled example: 2 events/turn but 8B parrots 0.53. Instruction placeholders: 8B 1.0 to 1.8 events/turn, parroting 0.
E3 (baseline, E3a-E3d)	How does the shipped pipeline behave, and did the v2 fixes help?	Linden DE+FR, 5 residents, 3 rounds, T=0; 70B n = 3 / 5, 8B n = 2 / 5	Baseline: 0/159 cited ids valid; 2/3 70B reports asserted a pass. Final: ids 1.00, 0/5 asserted, wall 126 s vs 334 s (unexplained). C1 corrected baseline self-intro to 0.27.
E4	Does the opinion update polarise residents with no conversation? (rule: [8, 9])	25 residents, 15 rounds, zero chats, 20 seeds, per controversy level	Confirmed: mean abs leaning 0.506 to 0.686 (medium) and 0.723 (high); 35.4% and 38.6% end above 0.9; low unchanged. v2: drift only for residents who conversed.
E5	What does retrieval actually hand the resident?	15 resident prompts of one baseline run; offline parse; six key facts	The vote question (118 to 124 %) reached only 4/15 prompts; other key facts 14-15/15. Language filter a no-op (all 9 chunks tagged de); chunks cut mid-word; corpus about 1.6k characters.
E6	Does the numeral-grounding gate handle edge cases?	12 hand-built cases against the Linden corpus, offline	Correct on only 8/12: flags correct derived arithmetic; hallucinated '2 Millionen CHF' and '4 Franken' pass via list numbers; spelled-out numbers missed. After v2: 10/12.
E7	How good is Apertus [61] on Swiss civic knowledge and Swiss orthography?	15 questions, 70B and 8B, regex rubric; ß test 3 samples x 4 prompts	70B 12/15, 8B 9/15. Both define Steuerfuss wrongly and fail 4 000 CHF x 6 % (70B 280, 8B 360; correct 240). ß per 1 000 characters: 70B 2.6, 8B 1.2.
E8	Does a silicon electorate [12] predict real Swiss federal votes [59]? (cf. [15, 16, 21])	54 votes 2021-2026, ballot title only, 70B and 8B, 3 672 calls	No: yes-bias +34 to +37 pp (70B, DE). Spearman 0.25 / 0.38 (70B), -0.02 / 0.02 (8B); Federal Council position alone 0.63; recalibrated MAE 12.4 pp vs 14.1 constant.
E9/E9b	Does the report assert invented outcomes or give vote advice?	70B, logged report prompt, 6 objectives x 5 samples = 30; replay with v2 template	Vote advice 0/30. Baseline asserted an invented outcome in 12/30 (lower bound; 4/5 for default and 'did it pass?'). v2 template: 0/30 asserted, German 30/30.
E10	What is the gateway's real parallel-request limit and throughput?	Raw requests, no retries, idle key, 1-8 in flight, both models	Effective limit is 4, not 5: at 5 in flight only 4/10 succeeded. 70B 156 tok/s at 4 in flight, 8B 555; app default 6 was too high.
E11/E11b/E11c	Do personas differentiate stance, and does a balanced corpus fix unanimity? (cf. [14, 32, 43])	15 residents from 70B baseline runs; T=0 and T=0.7; persona ablations; balanced corpus	Negative: 15/0/0 yes with full persona, role-only and no persona. Balanced corpus 14/0/0 (+1 failed call), T=0.7 27 yes, 2 no. Impact judged benefit for 14/15.
E12	Does long-context needle recall hold up to about 105k tokens?	70B and 8B, DE and FR, 5 lengths x 3 depths = 60 calls	60/60 correct. 70B 31-36 s per call at about 100k tokens, 8B 8-9 s. Single easy needle only; persona dilution untested.
E13/E13b	Does the LLM agree with the code-owned stance?	15 residents (v2.1), then 25 residents (final); re-asked with own memories	Agreement 0.40 in v2.1 (code no 4, LLM no 1); 0.64 in final. 7 of 10 code-against residents turn undecided or yes.
E14/E14b	Does resident dialogue tilt toward yes, and match the code stance?	238 utterances (E14), 465 lines (E14b); two LLM classifiers, 70B and 8B	No yes tilt: v2.1 70B dialogue 0.54-0.61 against vs 0.09-0.16 for. Held-for residents sounded for 17-42%. Final speech match 0.91 / 0.88 (70B standard).
E15/E15b	Can speech be bound to the code stance, and does prompt position matter?	Replay of 75 prompts, 3 conditions (E15); in-loop 3 x 70B, 2 x 8B (E15b)	Reminder as last paragraph best (0.84 / 0.85, 70B). In loop: 70B 0.46-0.55 to 0.80, 8B 0.50-0.52 to 0.73-0.76. Undecided did not improve (70B 0.53 to 0.58).
E16/E16b	Can an undecided resident sound undecided?	Replay of 41 undecided prompts, 70B and 8B; U0-U3, then U1p, U1h	U1 raised mixed 0.47 to 0.72 but copied arguments verbatim 62-67%. U1p: 70B mixed 0.74 / 0.72, copying 0.05; 8B mixed+neutral 0.53 to 0.68. Opening phrase and hints failed.
E17	Does the pipeline work on a real federal booklet [60]?	28 Sept 2025 booklet, 83k characters, 209 chunks, 14 questions, 70B, hand-graded	recall@4 10/14 (v1) vs 11/14 (v2, not significant); QA correct 11/14 and 12/14; unanswerable 3/3 abstained. Retrieval is the bottleneck; French untested.
E18/E18b	Does a clean clone build and run through the real UI?	Fresh clone, docker compose up, repo's Playwright script against UI and API	Failed at first: frontend image build error (lockfile without Linux packages), type error, concurrency default 6, swarm default, POST /simulate HTTP 422 (4 000-character cap). All fixed.
E19	Can the harness compare what-if conditions such as a council recommendation?	8B, 4 conditions x 3 paired seeds = 12 runs, standard graph	Yes or No recommendation shifts stance small and inconsistently (ranges include zero). Balanced corpus lowers stance in all 3 seeds (paired end -0.28). n = 3, 8B only.
E20	Which models and capabilities does CSCS offer compared with livemap?	n = 5 per cell (3 for probes); four Apertus models; livemap same day	Key reaches only four Apertus v1.5 models (others 403); no embedding model. Same tool and thinking limits; only 8B-thinking fills the reasoning field. json_schema, seed, logprobs work on livemap too.
E21	Does CSCS have a rate limit, and what throughput does it give?	1-96 in flight, short and 2 400-token prompts, 250 tokens out, no retries	0 non-200 in 2 496 requests; 70B 66 to 4 359 tok/s at 96 in flight. Livemap ceiling 5 in flight. Long jobs gave HTTP 504, notably on 8B-thinking.
E22	Does long context work up to 233k tokens on CSCS?	80 calls, DE and FR haystacks, 48k-233k tokens, single and three-fact questions	Single needle: 8B 30/30, 70B 26/30 (DE 11/15, digit corruption). Three facts: 70B 9/10, 8B 5/10. Livemap control n = 1 per cell, 2-3x slower.

E23	Is temperature 0 reproducible (cf. [43]), and does strict json_schema work on real prompts?	24 resident prompts x 3 repeats, T=0, both endpoints and sizes	Not reproducible: 2.5-2.75 distinct outputs of 3; seed no effect. json_schema on CSCS 8B valid only 11/24 (endless whitespace), 70B 23/24, livemap 24/24; shipped json_object 72/72.
E24	Is the yes-bias a property of the weights or of the deployment?	15 residents, Linden vote, T=0, four CSCS models, logprob view	Bias persists: 14 yes / 1 no with persona, 15 yes without, on 70B, 8B, 70B-thinking. 8B-thinking had unparsed answers (6 of 30 hit token limit). 70B P(yes) 0.93.
E25	Does the grounded 1:1 chat answer correctly without vote advice?	5 residents x 5 scripted questions, 25 answers per run, backend on CSCS, 3 iterations	Run 1: 8 flagged figures, undecided residents repeated their position. Final: 0 errors, 0 invented figures, unknown fact declined 5/5, 0/5 vote recommendations, fact cited 5/5.
E26	Is stuffing the whole booklet better than retrieval?	83k-character booklet, 14+3 questions, 70B and 8B, top 4, top 8, whole; hand graded	70B correct 11 (top 4), 12 (top 8), 13 (whole); 8B 11, 12, 12. Unanswerable 3/3 abstained; 94% of booklet tokens cached. Stuffing loses citation chips.
E27	Do the v2 fixes replicate on CSCS with the final code?	5 seeds x 2 sizes plus one 25-resident town; same code and seeds as livemap	70B wall 61 s vs 126 s livemap; 8B 23 s vs 28 s. Cited ids 1.00. Speech match 0.85 / 0.87 (70B) vs 0.91 / 0.88. 25 residents: 105 s, 200 calls, 163 events.
E28	Does the real booklet work through the whole app (PDF, simulation, chat)?	48-page booklet via POST /context/sources, 5 residents, 2 rounds, 25 chat answers	95 s, 0 errors. Retrieval alone: lost-revenue question 'not in the text' 5/5. Whole text in prompt: 5/5 answered 1.8 billion CHF, 0 invented figures, but only 2/5 carry a citation chip.
E29	Is Python the bottleneck of the simulation?	One instrumented 5-resident, 3-round run per size on CSCS; micro-benchmarks	Process CPU 2.2% (70B, wall 60.5 s) and 5.0% (8B, wall 26.1 s). Output tokens explain 81% of mean latency. Booklet retrieval 12.1 ms to 2.5 ms after caching.
E30	How does our retrieval compare with BM25 [62], dense [63] and hybrid [64] baselines?	14 gold questions, real booklet, 209 chunks of 480 characters, CPU fastembed	recall@4: BM25 as shipped 0.79, hybrid BM25 + e5-large 1.00, MiniLM dense 0.71 (worse than BM25). n = 14, one-question differences are noise. Option off by default.
E31	Can one build run on both endpoints, and does the limiter adapt?	25 residents x 2 rounds, both endpoints, LLM_CONCURRENCY=auto, 3 limiter iterations	Both id directions: 0 errors. Final limiter: CSCS 95-97 s, gateway 654 s (656 s at limit 4): more in flight added latency, not throughput. CSCS 6.7-7.3x faster.
E32	Can we count tokens, show metrics and stream the chat safely?	25 residents, 2 rounds on CSCS; 25 streamed chat answers	143 calls, 168 k tokens in, 51 k out, 16% cached, p95 14.4 s, 0 retries. Chat first text 1.5 s (median), whole answer 2.4 s; quality unchanged.

5.6 Development log

5 Oct: audit of the inherited code, instrumented baseline, probes (E1–E12), the yes-bias (E11) and the **v2 design** (E14–E19). **6 Oct:** CSCS endpoint (E20–E28), grounded chat. **7 Oct:** engineering questions (E29–E32); defaults switched to CSCS, which the judges use. **8 Oct:** literature and deployment survey; demo run recorded. **13 Oct:** clean-clone build; a deployment bug found while recording (two server workers, so about every other run failed with "Simulation not found") and fixed with one worker; scenarios, videos, this report.

6. Limitations

- **Not a vote predictor** (54 real votes; weak signal after bias correction; plausible output is not valid output [37], and replacing people with simulations undermines inclusion [34]). Stance weights are modelling assumptions, not calibrated; the stance prior *produces* disagreement and is not evidence that real residents would split that way. Evaluation uses one synthetic town text and one real booklet. LLM agent simulations need robustness audits before any scientific claim [36], which our repeated seeds and published defects partly provide.
- Small n: 3–5 simulations per group, 15 residents in population tests, 14 retrieval questions, 15 held-out votes; ranges, no significance tests.
- Two hosted endpoints over a few days (latency drifts several-fold); **local and air-gapped deployment were not measured** (no GPU), and the browser page loads one web font from Google Fonts, which an air-gapped installation must bundle. The native-speaker review of the German and French lines is prepared but not done.
- Open: the `price_pressure` indicator is always 0; the 8B never emits `move`; residents are capped at 25; the Linden sample is synthetic and short; the real-booklet French edition is untested.

7. Reproducibility

Hardware and runtime. Developed and measured on a Windows 11 laptop (Intel Core i5-1235U, 12 threads, no GPU) with Docker 29.8.1; the containers run Python 3.12 (backend) and Node (frontend). All inference was remote (Apertus 1.5 70B and 8B on the CSCS inference API and the hackathon gateway); the app needs no GPU. **Seeds.** The app draws residents and the stance prior at random for each run and has no seed setting, and `temperature 0` is not reproducible on Apertus (E1, E23), so two runs differ; the experiment scripts take a seed and log every call, and ranges over seeds are the result (the spread run shows them in the app); the recorded demo run replays identically. **Commit.** The results come from branch `main` at submission (last measurement commit `2e0bb72`, deployment fix `d5ed09a`); `git rev-parse HEAD` in a fresh clone prints the exact commit.

Single run command. Clone the public repository, set three environment variables (`LLM_BASE_URL`, `LLM_NAME`, `LLM_API_KEY`; the defaults are the CSCS endpoint and `swiss-ai/Apertus-v1.5-70B`) and run `make run` (without `make`: `docker compose up --build` in the project directory). The app opens at `http://localhost:3000` (a second port is `UI_ALT_PORT`). **LOAD** replays the recorded demo run without calling the model; **PLAY** opens the editor. The repository README lists **ready-to-paste scenarios** in German, French and English (one deliberately ambiguous), with what to look for, and a setup guide for the variables. 146 offline tests run with `uv run pytest` in the backend; every experiment is a script with raw call logs.

8. Next steps

Run the probe and a simulation on a local vLLM (8B); apply the native-speaker corrections; statistics-based personas; calibrate the stance prior on municipal results; a panel-question output so a human citizen panel (cf. Sion 2019) can check what the simulation flags.

License and references

License. CC-BY-4.0 (all Hack Apertus projects are open-sourced); the code is Apache-2.0. These pages are outside the six-page limit. Vendor and press items in the list are marked as such.

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